

OTOG QI, China

WMO: 535290

Lat: 39.0883N

Lon: 107.9625E

Elev: 4531

StdP: 12.44

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.7	-1.4	-23.0	1.8	8.1	-19.0	2.3	10.5	22.5	16.8	18.9	20.3	2.3	20	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	89.4	60.6	87.1	60.3	84.6	59.7	68.5	78.4	66.9	77.1	65.4	75.6	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	113.7	72.2	64.0	105.7	71.4	61.8	97.8	70.5	35.8	78.4	34.3	77.3	32.9	76.0	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.3	17.0	14.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-12.2	93.8	4.9	2.7	-15.7	95.8	-18.6	97.4	-21.3	98.9	-24.9	100.9		
			-12.9	70.3	4.7	1.9	-16.3	71.7	-19.0	72.8	-21.7	73.8	-25.1	75.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.7	15.9	23.5	36.2	50.1	60.7	69.3	73.3	70.0	60.2	47.5	32.4	19.3	(6)
(7)		DBStd	21.10	7.72	8.88	9.11	8.54	6.77	5.26	4.60	5.07	6.24	7.36	8.13	7.66	(7)
(8)		HDD50	3961	1056	743	435	102	7	0	0	5	133	528	952		(8)
(9)		HDD65	7333	1521	1163	893	450	166	20	2	165	543	978	1417		(9)
(10)		CDD50	2739	0	0	7	105	337	580	722	621	312	55	0	0	(10)
(11)		CDD65	636	0	0	0	4	32	150	258	170	22	0	0	0	(11)
(12)		CDH74	6236	0	0	1	74	474	1581	2437	1430	234	5	0	0	(12)
(13)		CDH80	1954	0	0	0	13	90	477	920	406	48	0	0	0	(13)
(14)	Wind	WSAvg	5.9	5.0	5.6	6.7	7.3	7.2	6.3	5.8	5.4	5.2	5.1	5.5	5.2	(14)
(15)	Precipitation	PrecAvg	10.3	0.1	0.1	0.3	0.4	0.9	1.0	2.4	3.2	1.5	0.5	0.2	0.0	(15)
(16)		PrecMax	17.6	0.4	0.4	1.1	2.6	3.3	2.5	6.0	9.8	5.0	2.0	0.8	0.4	(16)
(17)		PrecMin	4.9	0.0	0.0	0.0	0.0	0.0	0.1	0.6	0.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	3.7	0.1	0.1	0.3	0.6	0.9	0.6	1.3	2.5	1.2	0.4	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	52.6	68.5	81.6	86.3	91.2	94.5	90.0	86.3	74.2	59.3	44.0	(19)
MCWB			30.4	35.7	45.6	52.1	54.1	57.7	64.0	62.1	55.6	51.4	41.6	31.4	(20)	
2%		DB	38.2	46.5	61.5	75.3	82.4	87.9	90.6	87.2	79.6	69.4	54.7	39.3	(21)	
		MCWB	27.6	32.8	41.6	48.5	53.1	57.5	62.0	62.1	56.5	50.0	39.3	29.0	(22)	
5%		DB	33.7	42.5	56.5	70.7	79.5	85.2	88.0	84.5	76.1	65.3	50.6	35.7	(23)	
		MCWB	24.7	30.6	38.9	47.0	52.3	57.1	61.9	62.1	55.8	47.9	37.3	26.8	(24)	
10%		DB	29.7	38.7	52.2	66.4	76.0	82.6	85.4	81.8	72.9	61.4	46.4	32.1	(25)	
		MCWB	22.4	28.8	36.6	44.9	50.8	56.6	61.5	61.2	55.3	46.0	35.4	24.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	37.4	47.1	55.0	59.3	65.7	71.1	70.6	66.3	55.6	44.7	32.5	(27)
MCD B			43.0	49.7	63.6	68.0	72.2	75.3	80.6	80.1	70.9	67.3	54.1	41.7	(28)	
2%		WB	27.7	34.1	42.9	51.9	56.7	63.0	68.9	68.7	62.3	52.8	41.3	29.6	(29)	
		MCD B	37.4	44.3	58.4	67.3	72.7	74.0	79.4	77.7	69.5	63.7	50.2	38.1	(30)	
5%		WB	25.1	31.5	39.9	49.4	54.8	61.4	67.2	67.1	59.9	50.5	38.8	27.3	(31)	
		MCD B	33.1	41.0	54.7	65.4	71.7	73.3	78.7	76.5	69.2	60.5	47.8	35.0	(32)	
10%		WB	22.5	29.2	37.5	47.0	53.2	59.9	65.8	65.5	58.0	48.3	36.4	25.0	(33)	
		MCD B	29.5	38.5	50.7	63.9	69.9	73.5	76.9	75.1	67.8	57.6	45.8	32.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.6	23.3	22.9	23.1	22.8	21.5	19.8	19.5	20.1	21.7	21.1	20.7	(35)
MCD BR			27.6	27.5	28.7	29.3	28.3	26.2	24.5	23.6	25.3	26.9	25.9	24.6	(36)	
MCWBR			19.5	17.5	15.8	13.8	11.1	8.0	7.0	7.2	9.2	12.3	14.7	16.9	(37)	
5% WB		MCD BR	26.5	25.2	25.6	23.7	22.8	19.4	17.7	17.3	17.5	22.2	22.3	23.1	(38)	
		MCWBR	18.8	16.4	14.6	12.1	10.2	6.8	7.2	7.4	8.6	11.6	13.8	16.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.380	0.424	0.468	0.493	0.521	0.492	0.495	0.471	0.419	0.349	0.315	(40)	
(41)		taud	2.118	1.980	1.897	1.833	1.839	1.837	1.990	1.959	1.944	1.998	2.106	2.164	(41)	
(42)		Ebn at Noon	261	258	260	257	253	246	252	247	243	241	247	254	(42)	
(43)		Edh at Noon	35	47	56	64	65	66	56	56	54	45	36	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	874	1171	1592	1921	2144	2074	1955	1773	1480	1275	960	795	(44)	
(45)		RadStd	88	77	103	129	107	148	85	86	104	79	87	65	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	-2.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page